LEF MODEL · CHAPTER 27

Chapter 27: The Research Map of the LEF Model (Part A) — the Early Earth, the Formation of the Moon and the SPA Basin (Topics 1–79)

Based on Paper 14 — 14. Research topics: What the model opens for other researchers (Divisions A–G: the early Earth, the body at the tip, the orbit, the maria, the craters, the poles and the SPA basin)

01 · THE MYSTERY

The mystery: how do you turn a whole theoretical model into 132 independent studies that can be carried out right away?

After the full theory of the **LEF (Lunar Elevation and Fission)** model was formulated in Papers 1–13 and all its quantitative requirements were defined, **Paper 14 (Research topics)** serves as a practical working guide for the scientific community. The paper does not perform new calculations and does not change any numerical value in the series; instead it gathers all the open questions from all 12 previous papers — together with 15 research proposals from the author's working logs — into **132 distinct research topics** organised in 11 divisions (A–K):

1. **The five sources of the topics:** (1) requirements R1–R18 from Paper 13 not yet fully tested; (2) studies defined with pre-set decision thresholds and not yet run; (3) calculations marked in the series as "to be completed"; (4) the "traffic-light" rows (orange and red) from the summary table of Paper 1; and (5) 15 questions from the working logs that have not yet received a paper of their own.

2 **Three levels of readiness:** each topic is classified as **public data**, which can be carried out at once on existing catalogues and maps (such as GRAIL, LOLA, Robbins 2019 and GEBCO), **new modelling**, which requires a three-dimensional simulation or a mathematical integration, or **new data**, which requires laboratory samples or a space mission.

3 **The scope of this chapter:** in this chapter we survey the first seven divisions (**Divisions A–G, topics 1–79**), which deal with the formation of the Moon and its surface — from the early Earth to the South Pole–Aitken basin (SPA).

02 · AN IMAGE

An image: an engineering master plan in which every bolt is an independent experiment

Four tangible images present the practical character of the first 79 topics in Divisions A–G:

- **A hydraulic lift at a pressure of 50–150 GPa with a pipe 1,000 km wide (Division A, topics 1–11):** to lift a lunar mass, and a pear weighing 8.8–12.1 times as much, to the corotation line without the pipe freezing, the viscous-flow equation depends on the fourth power of the pipe's radius (b^4). The studies here examine whether the deep melt rich in water and CO_2 is lighter than the buoyancy threshold (3,864 kg/m³) and whether the sinking of iron into the core (~ 10^{31} J) supplies all the energy required (~ 9×10^{30} J).
- **Gluing two plates with a layer of molten honey — Stefan adhesion (Division B, topics 12–23):** the interface between the Moon and the head of the pear acted as a partly molten layer in which **Stefan adhesion** operated, until a fast breakout in the gap $d = 3d_0$ released the Moon ~ 210 km beyond the corotation line (17,496 km), while the slow pressure in the contact zone left dense minerals (coesite, stishovite and jadeite) in the Apollo 14 samples at Fra Mauro (topic 22)!
- **A boiling pot of soup spattering drops onto a cold lid and creating "steam caves" (Division D, topics 30–43):** the 34 maria are ejection openings that were sealed only when the melt flux dropped below the critical flux; and the pits and caves on the Moon (such as the cave beneath the Mare

Tranquillitatis pit) are proposed as **deep, branching steam cavities** that broke out, and not only shallow lava tubes (topic 38).

- **Controlled cannon fire in a steam atmosphere, and impact glass that keeps the fingerprint (Division E, topics 44–57):** about **99% of the Moon's craters** were formed by returning terrestrial material that crossed a steam atmosphere (10–1,000 bar); since the landing blew aside the thin dust coating, the chemical signature of the impactor must be read **inside the glass and melt created by the landing inside the crater** (topic 49 — one of the model's two key observations!).

03 · SCIENCE AND MODEL

Science and model: summary of the first seven divisions (Divisions A–G, topics 1–79)

Each division in the first part of Paper 14 gathers the numerical values, test methods and decision targets of a different stage of the Moon's story:

1. **Division A — the Earth before and during the pear (topics 1–11):** thermal expansion of the hot parent ($\Delta R = 120$ km, range 98–520 km); the layered Maclaurin problem ($D = 2 \implies 16,300$–$16,400$ km); core differentiation and the Hf–W clock over 600 Myr; the equation of state at 50–150 GPa; the energy of the lifting; the pressure test with the flattening ($J_2 \simeq 0.036$); the rate of build-up; the equilibrium shape of the pear (R9); the northward tilt to latitude 22.5°N; the pull on the Earth's core by 270–370 km and its return; and a fresh calculation of the classical fission path (2.74× against 1.007× today's angular momentum).

2. **Division B — the body at the tip: shape, rounding and release (topics 12–23):** a three-dimensional SPH simulation of the dome (3,650 km, contact shrinking from 6,000 to 3,000 km); a stiff shell ($\eta \gtrsim 10^{23}$ Pa·s) over a mobile base; the transition from a flattening of a/c ≈ 1.20 to a stiff, round body ($\mu \geq 10$ GPa, axis ratio ≤ 1.04); latitude drift; Stefan adhesion ($d_0 \approx 105$ km, margin ~ 210 km); the pressure trapped at detachment; survival inside the fluid Roche limit (17,496 km between 10,314 and 19,971 km); the layering of iron in the far crust; the melting of the sial in the contact zone; KREEP as the

folded edges of the sial (R12); high-pressure phases at Apollo 14; and the Moon's porosity by depth.

3. **Division C — the orbit (topics 24–29):** a coupled thermal-orbital integration from $r_0 = 17,496$ km and inclination $i_0 = 22.5^\circ$ to 384,400 km and 5.145°; the history of tidal dissipation (Q ≃ 12 today against an average of Q ≃ 29 over 3.87 Gyr); the start of recession on the first orbit ($\langle \dot{a} \rangle > 0$); the time of solidification of the one-sided magma ocean $20R_E$ ago; the frozen shape (J_2, C_{22}) from the closing of the dome; and reconstructing the 22.5° latitude without fitting to the orbit.

4. **Division D — the maria and the openings (topics 30–43):** the critical flux and the sealing times (topic 30 — the only one requiring a new model, while the other 13 are available at once on public data!); the order of sealing from the edges to the contact centre; a thorium minimum above the feeding channels; crater rings around volatile-rich maria; gradual exposure within a single mare; the difference between exposing openings and covered conduits; the titanium part of Study 13; the collar of the far-side maria (207 m); the pits and caves as steam cavities; the "crustal jump" (median 17 km) by basin age; opening diameter by feeding duration; the layered margins; blunt against sharp peaks at the edges of the plains; and the gravity traces of craters swallowed by active maria.

5. **Divisions E, F and G — the craters, the poles and the SPA basin (topics 44–79):** the exit speed (2–8 km/s) and the passage through the steam atmosphere (0.1–10 km); shock in a layered target at g = 8.25 m/s²; identifying the returning body in the crater melt (topic 49); a crater excess on the trailing side (1.96× / 1.45×); age gaps between crater counting and radiometric dating; the rays of young craters as a record within the ground; a uniform distribution of young craters at the poles (0.9–1.1) against the ancient density (1.20 ± 0.05); and the energy balance, the reduction ($\Delta \mathrm{IW} - 1.93$), the evaporation of metals (K, Na, Zn, Cu, Fe, Cl) and the closing of pores (2.9%–4.4%) in the SPA basin.

The mechanism in words: from the depths to the rim of the crater — the four pillars of topics 1–79

Divisions A–G of Paper 14 lay out a full physical sequence linking the depths of the early Earth with the detailed geology of the Moon's surface:

1. **The initial conditions in the Earth and the transfer of mass (Divisions A–B, topics 1–23):** a hot, volatile-rich parent body ($\Delta R = 120$ km, a day of 5.5 hours) pushes a melt lighter than 3,864 kg/m³ through a wide passage (b ~ 1,000 km, flow $\propto b^4$). The dome shrinks from a contact of 6,000 km to 3,000 km, stiffens to a shear modulus $\mu \geq 10$ GPa (axis ratio ≤ 1.04), and is released by the breaking of Stefan adhesion ($\Delta r \simeq 2d_0 \approx 210$ km beyond the corotation line, at 17,496 km) inside the fluid Roche limit (19,971 km) and outside the rigid one (10,314 km).

2. **The orbit, the maria and the returning bombardment (Divisions C–E, topics 24–57):** from the initial orbit ($i_0 = 22.5°$, $e_0 \simeq 0.037$) the Moon recedes to 384,400 km (i = 5.145°); at the same time, the 34 maria are exposed and sealed in the order of the critical flux, while terrestrial material ejected at 2–8 km/s crosses a steam atmosphere (10–1,000 bar) and creates ~ 99% of the Moon's craters — which explains why crater counting at a constant flux overstates the age of units compared with radiometric dating (such as Chang'e-6: 2.807 Gyr in the laboratory against 2.98 Gyr by counting, topic 56).

3. **The model's second key observation — reading the crater melt (topic 49):** while the ground around a mare carries an iron-rich coating (6.9 → 3.1 wt% FeO within 150 km against 2.3 wt% in the control), the impact itself blows the thin coating aside, so the crater looks like its surroundings. To identify the returning terrestrial body, **the melt and glass created by the landing inside the crater** must be sampled and compared with the source mare.

4. **The SPA basin as a thermal burner (Divisions F–G, topics 58–79):** the heat conduit that passed beneath the far side lifted a dome, closed the pores at a depth of 10 km to 2.9%–4.4% (against 6.2%–12.8% around), created

depolarisation and anomalous reduction ($\Delta\mathrm{IW} = -1.93$ in the Chang'e-6 samples against −0.80 on the near side), and evaporated volatile metals (K, Na, Zn, Cu, Fe, Cl), leaving a low K/Th ratio (451–455) and heavy isotopes.

05 · THE KEY EQUATIONS

Academic explanation and the key equations

The parameters that set the buoyancy threshold at deep pressure, the dependence of the flow rate on the passage radius b, the detachment condition of Stefan adhesion and the initial recession rate are formulated as follows (Divisions A–D):

$$\rho_{\mathrm{melt}}(50\text{–}150\ \mathrm{GPa}) < 3,864\ \mathrm{kg/m^3}$$

$$Q_{\mathrm{flux}} \propto b^4 \quad (b \sim 1,000\ \mathrm{km})$$

$$\Delta r_{\mathrm{release}} \simeq 2d_0 \approx 210\ \mathrm{km} \quad (d = 3d_0)$$

$$\langle \dot{a} \rangle > 0$$

These values link directly the micro-physics of water-rich melts at 50–150 GPa with the boundary conditions of the lunar orbit at the moment of release ($r_0 = 17,496$ km).

06 · THE SIGNATURE

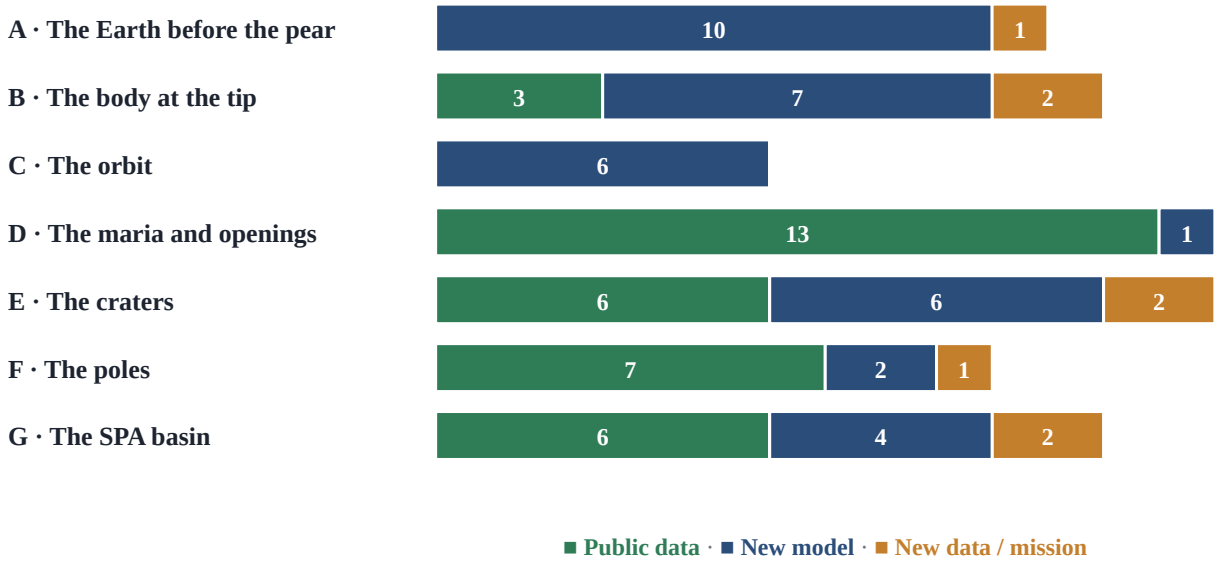
The signature in reality: selected examples of immediate decisive experiments in Divisions A–G

Of the first 79 topics, Paper 14 highlights several sharp experiments with an especially high power to decide:

- **Slow high-pressure phases in the Apollo 14 samples (topic 22):** the Apollo 14 landing site at Fra Mauro (3.6°S, 17.5°W) lies inside the contact dome (29°

from the contact centre at 24.1°N, 335.2°E), so its samples are expected to contain crystalline high-pressure minerals (coesite, stishovite, jadeite) formed under prolonged slow pressure — unlike the Apollo 16 and 17 samples (51°, in the edge belt), which carry only shock glass (maskelynite).

- **The order of sealing of the maria and the steam pits (topics 31 and 38):** a regression of the youngest basalt age in each mare as a function of the angular distance from the contact centre (holding the thorium concentration fixed), and checking for caves and pits on the far side far from any basalt as proof of deep steam cavities.
- **Young craters at the poles and the Serenitatis null experiment (topics 58 and 73):** counting ~ 150 young craters at the poles is expected to give a uniform ratio (~ 1.0) against 1.20 ± 0.05 in the ancient population, and a permutation test over all 34 openings will check how rare the coincidence is between the SPA minimum and the antipode of Serenitatis (197.9° against 198.8°).



Research topics 1–79 by division and level of readiness: green can be carried out at once on public data.

The lab box: mapping the first 79 research topics of the LEF model — Divisions A–G (Paper 14)

Division	Central subject	Range of topics (total)	Public data	New model / simulation	Mission / samples / laboratory
A	The Earth before and during the pear	1–11 (11)	0	**10** (1–3, 5–11)	1 (topic 4: equation of state at 50–150 GPa)
B	The body at the tip: shape, rounding and release	12–23 (12)	3 (19, 21, 23)	**7** (12–15, 17–18, 20)	2 (16 Stefan adhesion; 22 Apollo 14)
C	The orbit and tidal evolution	24–29 (6)	0	**6** (24–29)	0
D	**The maria and the openings (the 34 feeding openings)**	**30–43 (14)**	**13 (31–43)**	**1 (30: critical flux)**	**0**
E	**The craters and their source (key observation 49)**	**44–57 (14)**	**6 (51, 53–57)**	**6 (44–46, 48, 50, 52)**	**2 (47; 49 the crater melt)**
F	**The poles and the compaction of the rounding**	**58–67 (10)**	**7 (58, 61–62, 64–67)**	**2 (59–60)**	**1 (63 catalogue of the polar domes)**
G	**The South Pole–Aitken basin (SPA)**	**68–79 (12)**	**6 (72–73, 75–78)**	**4 (68–70, 74)**	**2 (71, 79 Chang'e-6 samples)**

07 · NEXT

The door ahead

In this chapter we mapped **the first 79 research topics (Divisions A–G)** of Paper 14 — from the physics of the early Earth and the body at the tip of the pear, through the orbit, the 34 maria and the craters (99% from returning terrestrial material), to the compaction of the poles and the thermal burner of the SPA basin.

In the next and closing chapter of the whole series — **Chapter 28: The Research Map of the LEF Model (Part B) — the Pacific, Mars, the Solar System and the Key**

Tests (Topics 80–132) — we will complete **Divisions H–K (topics 80–132)**, gather **the three selection lists of Part L** (the two decisive key observations, all the immediate topics on public data, and the tasks for simulations and for the MMX mission), and close with the two large fields of research outside the list: the Earth–Moon axis system and the magnetism of the Moon!

Chapter 28

The Research Map of the LEF Model (Part B) — the Pacific, Mars, the Solar System and the Key Tests (Topics 80–132) →

← Chapter 26 · Prologue

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